F O R M 2

THE PATENTS ACT, 1970
(39 of 1970)
And
THE PATENTS RULES 2003

COMPLETE SPECIFICATION
(See Section 10; Rule 13)

Short-range communication based automatic physical disconnection charger

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THE FOLLOWING SPECIFICATION PARTICULARLY DESCRIBES THE INVENTION AND THE MANNER IN WHICH IT IS TO BE PERFORMED:

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TITLE OF INVENTION

Short-range communication based automatic physical disconnection charger

FIELD OF INVENTION

The present invention generally relates to the field of a charger, particularly to an automatic disconnection charger. More particularly, the present invention relates to a short-range communication based automatic disconnection charger.

BACKGROUND OF INVENTION

KR20180026639A - it relates to a smart charger which is physically and automatically separated when charging is completed. According to the smart charger of the present invention, a lifespan of a charging battery can be extended, an electric accident due to excessive charging can be prevented, and an appropriate amount of electricity is used to reduce power consumption, and thus, an existing problem of continuously attempting to perform charging even when the charging is completed, is solved. The smart charger of the present invention comprises: a charging connection unit formed with 5-pin and 8-pin connection terminals for an Android smartphone to be connected to a smartphone and charged, in a general smart product charger; an automatic charging separator having an arm to extend to allow connection of a charging terminal to be separated, so as to physically separate a charging battery and a charger when the charging is complete; a current flow measurement sensor formed by a semiconductor sensor to measure a current flow; and a socket power connection unit including a cable and a socket connection terminal to be connected to an electric socket to receive power. According to the smart charger of the present invention, a lifespan of a charging battery can be extended, an electric accident due to excessive charging can be prevented, and an appropriate amount of electricity is used to reduce power consumption, and thus, an existing problem of continuously attempting to perform charging even when the charging is completed, is solved."

None of the above-mentioned prior arts neither teaches nor discloses about a short-range communication based automatic disconnection charger.

The invention in KR20180026639A uses a current sensor and a mechanical separator to physically disconnect the charger once charging is complete. In contrast, MagnoCharge relies on Bluetooth-based communication between the device and a microcontroller to control charging intelligently. It uses a mobile app to monitor battery levels and trigger disconnection at customizable thresholds. Instead of mechanical movement, it employs an electromagnet that deactivates to release the charging pin. Additionally, a magnetic reed sensor ensures the system activates only when the device is nearby, making MagnoCharge more compact, energy-efficient, and software-driven compared to the hardware-dependent approach in KR20180026639A.

OBJECTS OF INVENTION

One or more of the problems of the conventional prior art may be overcome by various embodiments of the system of present invention.

It is the primary object of the present invention is a short-range communication based automatic disconnection charger.

SUMMARY OF INVENTION

It is an aspect of the present invention is a short-range communication based automatic disconnection physical charger, comprising of;

A Bluetooth module;
A microcontroller;
An AC to DC Conversion Mechanism;
A custom printed circuit board;
A Functionally Modified Female USB;
A plurality of an outer pins;
A plurality of an inner pins;
An USB Connector;
A magnetic charging tip;
A metallic facing;
A power source; and
A miniature electromagnetic coil,

Characterized that,

The Bluetooth module (1) enables wireless communication between the adapter and the mobile device or external controller and it allows users to send commands to activate or deactivate the charging process remotely, therein the microcontroller (2) is the brain of the system, responsible for processing input signals, executing logic to control the charging mechanism, and interfacing with the Bluetooth module to receive user commands;

wherein, the AC to DC conversion mechanism (3) is responsible for converting the AC power from the wall socket into the required DC voltage suitable for the electromagnet and charging circuit and it ensures stable power delivery to the other components, therein the custom printed circuit board (PCB) (4) serves as the foundation for all electronic components, providing electrical connections between modules and ensuring a compact and organized layout and the functionally modified female USB (5) has the outer pins (6) (normally VCC and GND) are used for power transmission, and the inner pins (7) are used to power the electromagnet and these pins are repurposed to provide power to the electromagnet for activation;

wherein, the USB connector (8) that plugs into the power source or adapter, providing electrical input to the device, therein the magnetic charging tip (9) is a detachable charging tip that connects to the mobile device's charging port and the metallic facing (10) of the pin is attracted by the electromagnet; and

wherein, the power source (Central Conductor) (11) core conductor that carries power through the centre of the electromagnet and the charging tip connects this power source to the metallic contact at the centre of the tip, allowing the phone to receive power, therein the miniature electromagnetic coil (12) generates a magnetic field upon activation, allowing the magnetic tip to firmly attach to the charging port, ensuring a stable charging connection.

BRIEF DESCRIPTION OF DRAWING

Figure 1: illustrates a short-range communication based automatic disconnection charger.

DETAILED DESCRIPTION OF THE INVENTION WITH REFERENCE TO THE ACCOMPANYING FIGURES

The present invention as herein described about a short-range communication based automatic disconnection charger.

The present invention is a modern charging system whose advanced design includes the intelligent physical disconnection of a charging device from itself once a desired charge level is attained. Unlike traditional charges, whose function relies on electrically switched components such as relays and MOSFETs. It uses physical disconnection based on Bluetooth communication and an electromagnet-based system. Complete isolation from the power source ensures total overcharging prevention and top-off charging, prolongs battery life, and increases the energy efficiency of operation. The charger unit contains an AC-to-DC converter circuit that is used to convert the AC power available into a real DC voltage suitable for device charging. A microcontroller accesses the whole power delivery and the Bluetooth communication and also the electromagnet activation. A magnetic reed sensor to detect the charging application magnetic field and trigger the electromagnet. The modified USB port retains VCC and GND as the outermost pins, and the D+ and D- pins are re-used to control the electromagnet. This charging cable comes with an electromagnet aligned above its charging pin, with the magnetic reed sensor just close enough to detect the presence of the device. This device that will act as a charging target by offering a charging port compatible with USB-C, Micro-USB or Lightning connectors also has an embedded magnet that could enhance the interaction with the magnetic reed sensor. There is an application or background service on this device that will monitor the battery level and perform all charge management tasks, including sending a disconnect signal at the right time. When charging, the charger unit is plugged to a power source and supplies DC voltage through modified USB cable. The magnetic reed sensor is deactivated until the magnetic field of the phone approaches it. The power supply to electromagnet occurs when the phone is plugged, and the reed sensor gets activated by the magnetic field of the phone's speaker magnet trigger to facilitate the locking of charging pin through magnetic attraction. The Bluetooth module in the charger will pair with the mobile device, enabling a battery monitoring system. The app continually monitors the battery level of the phone while

it is charging and establishes communication with the microcontroller inside the charger. During this interaction with the application, the electromagnet stays ON and exhibits a secured connection between the charger and the mobile device. When the battery gets to the correct charge threshold, say 80 or 90 percent, the application will disconnect it from the charger via Bluetooth through a command sent to the microcontroller, which processes the command and turns off the electromagnet. When the magnetic attraction ceases, the charging pin automatically detaches from the charging port of the device, leaving it completely isolated and the residual power flow is eliminated. Since the charging pin is no more attached, no power from the charger is drawn, and the system goes to sleep mode. This resets the magnetic reed sensor, which means that the electromagnet will not be reactivated until the phone's magnetic field is again detected. The charger remains under standby with very little or even no power wastage. It is far better than conventional charging methods. For example, a normal charger will continue trickle charging the battery even after its full charge completion, whereas it cuts off completely as soon as the charging is complete. Moreover, it removes battery wear, as the charging pin is physically separated and no more residual current will flow through the device, thus giving well-being and longer life to the battery. This integration of the magnetic reed sensor thus makes the system an energy efficient and eco-friendly solution because it eliminates unnecessary power consumption. Complete prevention of trickle discharging makes it a great saver against overheating, short-circuiting, and swelling in the batteries, thus assured safety. The system is compatible with a wide variety of devices such as smartphones, laptops, electric vehicles, medical devices, and IoT gadgets, which makes it a versatile and scalable charging solution. It is a revolutionary charging system as it has a method of disconnection based on electromagnet, which is totally managed through Bluetooth, thus improving safety, battery life, and energy efficiency. It is state-of-the-art technology as it renders smart charging applicable to multiple applications by removing trickle charging, residual power, and energy wastage with complete physical disconnection after the required charge level is achieved.

The embodiment having simplified the design by using the DC output of an existing charger adapter instead of implementing an AC to DC conversion mechanism. The DC output is supplied to a circuit board that includes an Arduino UNO, which acts as the microcontroller, and a Bluetooth module to receive signals from the device when

the battery reaches the preset level. Since, the 5V output of the adapter is insufficient to power the electromagnet, an amplifier is incorporated to boost the voltage. It has also made use of 4 pins conducting pins which replicate the USB functionality. Two of these pins are used for charging the device and the other 2 pins are used for powering up the electromagnet. In addition, the present invention is as an application using MIT App Inventor, a popular platform for creating Android apps without writing code, making the development process more accessible. The app monitors the device's battery level every second. When the battery reaches the threshold level, a signal is sent via Bluetooth to the charger. For this functionality to work, the charger's Bluetooth module and the mobile phone must be connected. An electromagnet coil is made by hand which can function as a magnet when power is supplied to it. Since, the electromagnet consumes some amount of Energy we have installed a magnetic reed meter which can detects the magnetic field of the mobile phone's speaker. When the magnetic reed sensor detects any external magnetic field i.e magnetic field generated by the magnets of the mobile phone speaker it activates the electromagnet otherwise it does not consume any power and does not behave like a magnet.

First the user needs to plug in the charger into the power source. The external charger adapter converts the AC input to DC output. Again, the DC output is feed to the circuit which contains microcontroller, Bluetooth Module and an Amplifier. The DC current will be given to the Microcontroller and the Bluetooth Module and both were powered on. On other side the user needs to open the Application installed on the mobile phone and needs to connect to the Bluetooth module of the charger. After connecting the user also needs to set a desired threshold level for which the charger will be physically disconnected. Upon setting the desired level, the user needs to show the mobile phone speaker face to the magnetic Reed sensor. Then the Reed sensor read the field and informs the microcontroller. Then it is the job of the Microcontroller to turn on the Amplifier and makes it supply power for the Electromagnet. The electromagnet then behaves as a magnet and attracts the pin. Then the circuit become closed and charges the device. Once the device starts charging, the app which is preinstalled will monitor the battery level for every one second. Once the battery is charged to a desired level then the application immediately sends a signal to the charger via Bluetooth. The Microcontroller makes the amplifier shutdown. Then

the electromagnet loses its magnetic property, then the pin will be released which makes the physical disconnection.

Specifically the embodiment works in the way that a Bluetooth module (1) enables wireless communication between the adapter and the mobile device or external controller and it allows users to send commands to activate or deactivate the charging process remotely, therein a microcontroller (2) is the brain of the system, responsible for processing input signals, executing logic to control the charging mechanism, and interfacing with the Bluetooth module to receive user commands. An AC to DC conversion mechanism (3) is responsible for converting the AC power from the wall socket into the required DC voltage suitable for the electromagnet and charging circuit and it ensures stable power delivery to the other components. A custom printed circuit board (PCB) (4) serves as the foundation for all electronic components, providing electrical connections between modules and ensuring a compact and organized layout and a functionally modified female USB (5) has the outer pins (6) (normally VCC and GND) are used for power transmission, and the inner pins (7) are used to power the electromagnet and these pins are repurposed to provide power to the electromagnet for activation. An USB connector (8) that plugs into the power source or adapter, providing electrical input to the device, therein the magnetic charging tip (9) is a detachable charging tip that connects to the mobile device's charging port and the metallic facing (10) of the pin is attracted by the electromagnet. A power source (Central Conductor) (11) core conductor that carries power through the centre of the electromagnet and the charging tip connects this power source to the metallic contact at the centre of the tip, allowing the phone to receive power, therein the electromagnetic coil (12) generates a magnetic field upon activation, allowing the magnetic tip to firmly attach to the charging port, ensuring a stable charging connection.

CLAIMS

We claim:

1. A short-range communication based automatic physical disconnection charger, comprising of;
   A Bluetooth module (1);
   A microcontroller (2);
   An AC to DC Conversion Mechanism (3);
   A custom printed circuit board (4);
   A Functionally Modified Female USB (5);
   A plurality of an outer pins (6);
   A plurality of an inner pins (7);
   An USB Connector (8);
   A magnetic charging tip (9);
   A metallic facing (10);
   A power source (11); and
   A miniature electromagnetic coil (12),

Characterized that,

The Bluetooth module (1) enables wireless communication between the adapter and the mobile device or external controller and it allows users to send commands to activate or deactivate the charging process remotely, therein the microcontroller (2) is the brain of the system, responsible for processing input signals, executing logic to control the charging mechanism, and interfacing with the Bluetooth module (1) to receive user commands;

wherein, the AC to DC conversion mechanism (3) is responsible for converting the AC power from the wall socket into the required DC voltage suitable for the electromagnet and charging circuit and it ensures stable power delivery to the other components, therein the custom printed circuit board (PCB) (4) serves as the foundation for all electronic components, providing electrical connections between modules and ensuring a compact and organized layout and the functionally modified female USB (5) has the outer pins (6) (normally VCC and GND) are used for power transmission, and the inner pins (7) are used to power the electromagnet and these pins are repurposed to provide power to the electromagnet for activation;

wherein, the USB connector (8) that plugs into the power source or adapter, providing electrical input to the device, therein the magnetic charging tip (9) is a detachable charging tip that connects to the mobile device's charging port and the metallic facing (10) of the pin is attracted by the electromagnet; and

wherein, the power source (Central Conductor) (11) core conductor that carries power through the centre of the electromagnet and the charging tip connects this power source to the metallic contact at the centre of the tip, allowing the phone to receive power, therein the miniature electromagnetic coil (12) generates a magnetic field upon activation, allowing the magnetic tip to firmly attach to the charging port, ensuring a stable charging connection.

ABSTRACT

The present invention is a short-range communication based automatic physical disconnection charger. A Bluetooth module (1) and a microcontroller (2) is the brain of the system, responsible for processing input signals, executing logic to control the charging mechanism. An AC to DC conversion mechanism (3) and a custom printed circuit board (4) serves as the foundation for all electronic components, and a functionally modified female USB (5) has an outer pins (6) are used for power transmission, and an inner pins (7) and an USB connector (8) that plugs into the power source or adapter, providing electrical input to the device. A magnetic charging tip (9) is a detachable charging tip that connects to the mobile device's charging port and a metallic facing (10) of the pin is attracted by the electromagnet. A power source (11) and a miniature electromagnetic coil (12) generates a magnetic field upon activation, allowing the magnetic tip to firmly attach to the charging port, ensuring a stable charging connection.